

Online Redo Logs

⇒ Definition on previous page.

Notes:

- Each online redo log group must contain online redo log members.

- The member is the physical file that exists on disk.

1. Protecting online redo logs:

- multiplex groups to have multiple members.

- Never allow two members of the same group on the same physical disk.

- Ensure that OS file permissions are set appropriately.

- Use physical storage devices that are redundant.

- Appropriately size the log files so that they switch and archived at regular intervals.

set the ARCHIVE LOG Target initialization parameters to ensure that the online redo logs are switched at regular intervals.

2. Displaying Online Redo-log information:

- SELECT

a.group#, a.member, b.status, b.archived, bytes/1024/1024 mbytes
FROM v\$logfile a, v\$log b
WHERE a.group# = b.group#
Order By 1, 2;

3. Determining the optimal size of Online-log Groups:

- SELECT count(*)

, to_char(first_time, 'YYYY-MM-DD:HH24')
from v\$log_history
group by to_char(first_time, 'YYYY-MM-DD:HH24')
Order by 2;

- SELECT optimal logfile size
FROM v\$instance_recovery

4. Avoiding Delays in switching online Redo log:

- o Add more redo-log groups.
- o Lower the value of FAST_START_MTTR TARGET. Doing so causes the db-writer process to write older modified blocks to disk in smaller time frame.
- o Tune the database writer process.

5. Adding online Redo-log groups:

- o ALTER database
ADD logfile
group 5
('02a01/02a02/011R2/redo05a. redo',
'02a02/02a02/011R2/redo05d. redo')
SIZE 500M;

6. Resizing online redo-log groups:

- o Add log groups with files sized appropriately.
- o Drop the old log groups.

7. Adding online redo-log files to a Group:

- o ALTER database
ADD logfile member
'02a02/02a02/011R2/redo04c. redo'
to group 4;

8. Removing online redo-log files from a group:

- o SELECT a.group#, a.member, b.status,
b.archived,
SUM (b.bytes) / 1024 / 1024 mbytes
FROM v\$logfile a, v\$log b
WHERE a.group# = b.group#
GROUP BY a.group#, a.member, b.status,
b.archived
ORDER BY 1, 2

- o ALTER database logfile member
'02a02/02a02/011R2/redo04c. redo';

9. Moving or Renaming redo-log files:

add New log files in the new location and drop the old log files.

- o Physically move the files and update the control file with new location.

1. Shut down database.
2. From the OS prompt, move the files.
3. Start up db in mount mode.
4. update the control file with the new file locations and names via the ALTER DATABASE statement.
5. Open the database.

